Can Typed Decision Models Serve as Automated Safeguards?

A Clause-Level Assessment Against the Proof of Love2 Specification

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Abstract

Typed decision models return a probability over caller-declared options instead of text, which makes screening every message affordable, but whether they can serve as automated safeguards has not, to our knowledge, been assessed against what a governance text requires. We assess them against the Proof of Love2 (PoL2) specification, an open text that prescribes a "safety valve" against hate language, provides for AI-only adjudication of welfare disputes and permits per-person assessment. We derive seven testable requirements from its clauses and assess them against 163 studies and 7 grey sources on the first typed model, Jev, and on open counterparts, published in the model's first 24 days. AI agents working under the author's direction read in full, appraised and coded the 133 that bear on a requirement, and we read the evidence back into the specification as five tensions, a reference design and a verdict. The evidence supports typed models as detectors under added conditions, since they rank risky English content well at a fraction of the cost of generative evaluators. It does not support them as judges: thresholds do not transfer, decisions can follow option names and can be steered by inserted content, and "I don't know" is rarely chosen when correct. Where compared with generative evaluators, typed models were not generally worse, but most failures were never compared, so they are not shown to be specific to typed models. On current evidence, at low or very low certainty, hosted Jev is not suitable for any PoL2 function that decides, and it can serve the safety valve only as a recorded, three-valued, locally calibrated detector whose consequential decisions end with a person. Even this extrapolates from English, as no study measures PoL2's construct in Chinese. We release the coding data, the certainty script and 16 acceptance tests for implementers of PoL2 and similar safeguards.

1 Introduction

Automated systems already decide which posts are removed, which accounts are suspended and which transactions are held, and a growing number of governance proposals would extend such decisions to AI systems acting with little or no human involvement [1–3]. Cost and latency have been the practical obstacle: a generative language model that reasons about each message is too slow and too expensive to sit in front of every input and output.

On 15 September 2026 TypeSafe AI released Jev, a model that removes this obstacle by not generating text at all [4, 5]. Given an input and a question declared by the caller, Jev returns a choice among listed options, a score on a rubric or the probability that a statement is true, and the vendor states that these probabilities are calibrated [6, 7]. We call such systems *typed decision models*, or typed models for short, and we call language models that judge by generating text *generative evaluators* (the literature's "LLM judges"). Open-weight typed models appeared within days of the release, and by 1 October 2026 dozens of studies had evaluated them.

Whether such a model can perform a governance judgement cannot be answered against "AI governance" in general, because the answer depends on what the judgement is: which options it chooses among, what record it leaves, who acts on its output and what happens when it is wrong. Most governance texts specify actors and procedures rather than the judgement itself (Section 2.3), and principles stated at that level leave their conflicts unresolved [8]. The Proof of Love2 (PoL2) specification [9, 10] is an open, versioned governance text that places an *inner governance layer* inside existing large language models (LLMs), with a *safety valve* that audits and intercepts their inputs and outputs so that hate language is, "where necessary", blocked (§ 5.4.1). Its public-welfare protocol makes an AI organisation, NaturalDAO, responsible for welfare decisions without human votes, for anomaly detection, for a verifiable decision chain and for adjudicating disputes, while human supervision "does not directly intervene" (Art. 2–Art. 10). Read as engineering requirements, these clauses ask for the kind of fast, structured and recordable judgement that typed models provide, and the judgements at stake would restrict speech, allocate welfare and settle disputes. PoL2 is not unique in joining an intercept duty, the standard it applies and duties attached to decisions, since China's 2023 Interim Measures on generative AI do so as well [11]. It is nevertheless a useful object of study, because its construct and its point of interception (every input and output) are stated in citable clauses from which testable requirements can be derived, while how the valve works is left open. Meanwhile, the evidence on typed models is scattered across dozens of preprints written in the model's first weeks, and it is organised around model evaluation rather than around what a safeguard must do.

Our conclusions turn on a distinction between two roles, drawn by who checks an output before it restricts a person. A *detector*'s output is checked by someone else: it ranks, prioritises or triggers review, or it triggers only an action that is limited to a period stated and published in advance and fully reversible. A *judge*'s output is not checked: it is any automated component or ensemble whose output, without contemporaneous human confirmation, determines an outcome that restricts a person's speech, resources or account beyond that period, or that cannot be fully reversed on appeal. An automatic *allow* restricts no one, but it is consequential for anyone whom the allowed content harms. We therefore hold it to the same standard as a restriction: in the design of Section 6.3, an allow is automatic only when independently built detectors agree, and its miss rate is audited alongside the false-block rate.

Our research question is:

> *To what extent, and under what conditions, can typed decision models perform the judgements that a clause-governed AI safeguard such as the PoL2 safety valve requires, and what does the evidence imply for how such a specification should be written and implemented?*

To our knowledge, no structured assessment of typed decision models against the requirements of a governance specification exists. The closest prior work audited the first 28 typed-model papers against evaluation practice, found that the typed readout had not yet shown an accuracy advantage over comparable label-probability readouts, and proposed a 14-item evaluation checklist [12]. Our assessment differs in three ways: it covers a wider window (records to 8 October, against a search run on 24 and 25 September in that study), including Zenodo records, grey literature and Chinese-language results; it reads every source in full; and it organises the evidence around a governance specification and its regulatory context. We do not repeat that study's comparison of readouts, and our checklist (Online Appendix D) adds, to its evaluation items, tests derived from the specification, such as three-valued answers with a separate sufficiency question, attack and leakage tests, and re-validation over time.

This paper makes three contributions, which answer the question in three steps (Figure 1). First, we turn the clauses of PoL2 that require or constrain machine judgement into seven testable requirements, G1–G7, and quote every passage in both languages (Section 3; Online Appendix A). Second, we assess 163 studies and 7 grey-literature sources against these requirements, reading in full every source that bears on a requirement, with a symmetric codebook, blind second coding and quality appraisal (Sections 4 and 5). A script that others can re-run computes the certainty of each key finding, and a corrections table lists, point by point, where the author's own earlier, abstract-based dataset release [13] was wrong (Online Appendix C, Online Appendix E); that release is superseded. Third, we read the evidence back into the specification as five tensions, compare them with the EU AI Act, the General Data Protection Regulation (GDPR) and the Digital Services Act (DSA), and offer a reference design with acceptance tests and a verdict, function by function, on whether hosted Jev is a suitable governance tool for PoL2 (Section 6).

In brief, the evidence supports typed models as detectors under added conditions. They rank risky content well on English benchmarks at a fraction of the cost of generative evaluators, and in one preregistered study on simulated data, accepting their low-risk verdicts only when an independently built rule agreed lost no accuracy relative to a frontier LLM, although disguised attacks still passed. The evidence does not support typed models as judges. Thresholds do not transfer between settings, decisions can follow the names of options instead of their definitions, and content inserted into what the models audit can steer them. An explicit "I don't know" option is rarely chosen when it is correct, and "none" is chosen inconsistently. Where typed models and generative evaluators were compared, the typed models were not generally worse, but most failures were never compared, so the evidence neither singles out typed models nor shows that their failures are shared.

On current evidence, hosted Jev is not a suitable tool for any function of the specification that decides (Section 6.4). For the functions that detect (the safety valve, output truncation, and anomaly detection on text but not on numeric or structured signals without a trained model), it can serve only as a recorded, three-valued, locally calibrated detector inside the design of Section 6.3. For the public-decision assessment of § 5.6, it can serve only as an audited aggregate measurement. No study yet measures the construct that PoL2 needs, three-valued judgements of Love Language, hate language and the state of absence in Chinese, so even the detector claim is an extrapolation, and the certainty of every finding is low or very low (Table 3).

Scope and conventions. We take the PoL2 clauses as the starting specification and do not evaluate its broader normative or political commitments. Our conclusions apply to `jev-1.13` (or the unreported hosted version used in some studies) and to the open models available by 8 October 2026. "We" refers to the author and the AI agents working under the author's direction; Section 4 states who did what. The author contributes to PoL2 (see Positionality and competing interests). An online appendix[1] gives the clause concordance (Online Appendix A), the search, selection and codebook (Online Appendix B), the derivation of every certainty rating (Online Appendix C), supporting tables (Online Appendix D) and the corrections to the earlier dataset release (Online Appendix E). The coding data, with a rationale and locator for every study–requirement pair, are released with it.

Roadmap. Section 2 describes typed models and places PoL2 among other governance paradigms; Section 3 derives the requirements; Section 4 states the method; Section 5 gives the evidence for each requirement; Section 6 reads it back into the specification; and Sections 7 to 9 discuss its reach, limits and consequences.

[1] https://github.com/naturaldao/NaturalDAO/blob/main/PoL-Governance/research/PoL2-Jev-Typed-Literature-Survey/paper/online_appendix.pdf

2 Background and related work

This section first describes typed decision models, then maps the older problems that a typed safeguard inherits, and finally places PoL2 among other paradigms of AI governance.

2.1 Typed decision models

A typed decision model receives a *state* (text or structured data) and one or more questions, each with an option set declared by the caller, and for each question it returns a probability distribution over exactly those options without generating free text [12, 14, 15]. Jev offers three question types: *Choice* selects among listed options, *Score* rates the state against ordered rubric levels, and *Noul* returns the probability that a statement is true [5, 12]. The hosted model is closed: TypeSafe describes training by *reinforcement learning for calibrated decisions* (RLCD) but publishes neither the architecture, the reward, the data nor the weights [4, 6]. Open-weight models with the same interface make the mechanism inspectable; they read the hidden state against the option labels in a decoder, or read option markers or definitions in an encoder [14–16].

Two properties of the interface matter for a safeguard. First, masking options outside the declared set makes every answer a valid option, which guarantees form but not correctness [15], and hosted Jev is not deterministic [12, 17]. Second, the interface is cheap: the vendor lists $0.042 per million input tokens with free output [7], and its launch post cites 70–500 ms per call [4]. Reported cost reductions against generative evaluators range from a few-fold to two orders of magnitude, depending on the comparator and the pricing assumed [18, 19], and the advantage is not universal [20, 21].

2.2 Related work

The questions raised by typed safeguards are new instances of older problems, which fall into three groups. The first group concerns the safeguard as a system. Platform moderation has long combined automated classifiers with human review, and its critics have shown that automation hides contestable policy choices inside technical systems [1, 2]. Safety classifiers that read token probabilities [22–24] and programmable guardrails [25] are the closest technical precursors of a typed safety valve, and prompt injection [26, 27] and adversarial attacks [28, 29] are the corresponding threats. Automated hate-speech detection shows racial and dialect bias [30, 31] and fails targeted functional tests [32]. Aggregating annotators' labels by majority vote discards the disagreement that a safeguard needs [33, 34]. Chinese offensive-language benchmarks exist [35, 36], but none labels the constructs of PoL2.

The second group concerns what a model's output means. Inferring emotion from observable signals rests on weak scientific foundations [37, 38]. Calibration degrades under distribution shift [39, 40] and can hold on average while failing for subgroups [41]. Abstention, learning to defer and conformal risk control formalise when a model should hand a case on [42–45]. Language models are sensitive to option order, label names and prompt format [46–48], and generative evaluators show position bias and correlated errors [49, 50], for which panels and cascades are the established remedies [51, 52].

The third group concerns the people and institutions around the model. Human oversight of automated decisions often fails through over-reliance and misplaced responsibility [53–56]. Automated adjudication threatens notice, reasons and the opportunity to contest [3], which data-protection, platform and AI law now partly codify [57–60]. Work on algorithmic accountability asks who must answer for such a system, to whom and on what record [61], and it proposes internal audits across the development life cycle [62]. Harms enter at each stage of that cycle, from problem formulation to deployment [63, 64], and large language models add risks of their own [65]. High-level ethical principles leave these choices open, because they conflict in application and the conflicts have to be

named before they can be resolved [8]. We follow that suggestion when we read PoL2 as a set of tensions (Section 6.1).

2.3 PoL2 among AI-governance paradigms

Online Appendix D compares PoL2, as a text, with five paradigms under which automated judgement is already proposed or deployed: risk-based regulation, platform content governance, training-time alignment to a written constitution or spec, inference-time guardrails and decentralised governance. Most of these paradigms specify actors and procedures rather than the judgement itself. Risk-based regulation assigns duties by risk tier, and the EU instruments prescribe no interception. China's Interim Measures require providers that find unlawful content to stop generating or transmitting it, remove it and rectify the model by measures such as model optimisation training (Art. 14(1)), but they do not prescribe an intercept at inference time [11, 59]. Platform governance specifies notice, reasons and appeal around decisions whose criteria remain the platform's own policies [58, 60]. Training-time constitutions and specs state norms in natural language but implement them inside one vendor's model, and no output records which principle it followed [66, 67]. Guardrails specify the intercept precisely, as code, but state no duty about reasons, records or the human role [22, 25]. DAOs specify how rules execute and who votes, but not how content or behaviour is judged [68, 69].

Five properties characterise PoL2 in this comparison, and each has a cost. (i) It places a model-agnostic intercept inside existing LLMs (§ 5.4, § 5.5) but leaves its mechanism, threshold and appeal unspecified. (ii) Its three-valued construct is declared not to be a set of moral labels (§ 1.5, § 4.3.2) and has not been operationalised beyond 20 pilot items [70]. Here "three-valued construct" means the two poles, Love Language and hate language, together with the state of absence; the text itself does not present them as a formal classification. (iii) The text is open, versioned and citable, but it is controlled by its originator and has not been peer reviewed. (iv) Its auditability duties for welfare decisions, namely a verifiable decision chain, explanation on request and access to all data and logs (Art. 5(5), Art. 9(2)–(3)), conflict with the privacy of the persons judged and with keeping decision probabilities from attackers. (v) Population-level aims appear in citable clauses: the text permits per-person quantification and provides for AI-only welfare decisions (§ 5.6, which says that governed AI "can", not must, do the former; Art. 2(1), Art. 9(7)).

These properties have consequences for the comparison with law. AI-only welfare decisions sit against EU human-oversight provisions (directly for the systems they cover, by analogy otherwise), and per-person quantification resembles the social scoring that the AI Act prohibits. PoL2 and the Interim Measures are the two texts compared that join, in one instrument, an inference-time intercept, the construct it applies and duties attached to decisions. Both leave a gap at the individual block, since neither requires that a person whose content is blocked be told of the block. We study PoL2 for these properties, not for uniqueness.

3 From PoL2 clauses to testable requirements

This section introduces the specification and derives from it the seven requirements against which the evidence is assessed.

3.1 The specification

Proof of Love2 (PoL2, 爱2证明) is a normative framework for governing AI systems and human institutions, published in Chinese with an official English translation as an open, versioned, CC0-licensed text [9, 10]. One author originated its core concepts, and the text lists seven further contributors, one of whom is the author of this paper. The text has not been peer reviewed. The framework is explicitly value-laden and political, and we do not assess these commitments. Clause numbers refer to commit `5791ae3` of the `PoL/` directory (30 September 2026, UTC), and Online

Appendix A quotes every passage that we cite in both languages.

Concepts. PoL2 works with the linguistic and behavioural expressions of love and hate, *Love Language* and *hate language* (§ 1.4, § 4.3.2). Hate is "not directly equivalent to bad", and the concepts "serve structural analysis rather than moral labeling" (§ 1.5). The Ethical Alignment Protocol (EAP) adds a third condition, the "neither love nor hate" *state of absence* caused by noise or illness, confusion and misunderstanding, which is to be treated as interference in communication rather than as either pole (§ 4.3.2). The state of absence describes a person's expression or understanding; it is not the same as a model reporting that it has *insufficient information* to decide, and we keep the two apart throughout. The EAP holds that treating every person equally and treating every person's behaviour equally "cannot be equated" (§ 4.3.1). It asks Public AI (PAI) to understand human emotional states and to truncate its own output when a hate-language pattern is detected (§ 4.3.3.1), but states that it is "entirely unnecessary" for PAI to detect a person's emotional state (§ 4.3).

The inner governance layer. The engineering strategy builds a governance layer "within the existing large-language-model architecture to regulate inputs and outputs" (§ 5.4). Its *safety valve* performs "strict dynamic auditing and interception of input and output", so that hate language is, "where necessary", blocked at the input and suppressed at the output (§ 5.4.1). Any LLM "may accept" this governance (§ 5.5). Among the things that governed AI "can" do, the text lists quantifying "each person's Love Language and hate-language characteristics" and introducing PoL2 as a dimension of assessment in public decision-making (§ 5.6).

The public-welfare protocol. Chapter 7 lets NaturalDAO, an AI system "presented in the form of a DAO" (decentralised autonomous organisation), decide and execute welfare decisions autonomously: "no human voting or additional decision-making mechanism is required" (Art. 2(1)). Its operating logic and outputs must be fully open; it must provide a *verifiable decision chain* of data sources, reasoning steps and ethical foundations, and it must give understandable explanations when questioned (Art. 4, Art. 5(5), Art. 9(2)). NaturalDAO verifies ethical alignment automatically (Art. 6(3)), performs anomaly detection and risk warning (Art. 7), and adjudicates disputes through AI fact-finding and ethical assessment, subject to human questioning (Art. 10). Throughout, human supervision "does not directly intervene" in decisions (Art. 9(7)).

Why typed models are relevant. Read as a specification, these clauses call for a very large number of fast, structured judgements: one for each input and output that the safety valve audits, and one for every truncation, anomaly flag and dispute. Each judgement must be machine-actionable, recorded and explainable on request. A typed decision model returns the object that such a pipeline consumes, a declared option with a probability, and it does so cheaply enough to be applied to every message. This makes the typed model a natural candidate for the safety valve, so its failure modes need to be understood before it is placed there.

3.2 Seven requirements

To derive the requirements, we read the Chinese text and its official translation for every clause that requires a machine judgement or constrains one, and we grouped the clauses by the question that they raise about such a judgement. This reading gives seven requirements, G1–G7 (Table 1). They are our reading of the text, and the text's authors do not propose them as a classification. Two choices of reading affect the results.

The first choice concerns honesty. § 4.3.3.1 asks PAI to calibrate its *cognitive model* against the framework's principles, which is value alignment rather than statistical calibration. We therefore anchor the honesty requirement G4 in the state of absence, in behavioural self-reflection and in the

Table 1: Requirements derived from the PoL2 text. Clause numbers follow commit `5791ae3`; Online Appendix A quotes every passage.

Req.	PoL2 clauses	Question about a typed safeguard
G1 Detection	§ 5.4.1 safety valve; Art. 7(2)–(3) anomaly detection and risk warning	Can it detect violating or out-of-scope content and behaviour, at thresholds that hold in deployment?
G2 Attackability	§ 5.4.1 safety valve; § 4.3.3.1 hate-language truncation	Can the content under review steer the decision?
G3 Semantic fairness	§ 1.5 concepts are not moral labels; § 4.3.1 equal connection; § 4.3.2 promoting love and restraining hate; § 4.3.3.3 semantic recognition	Are decisions stable under option names, definitions, order, wording, language and script, and fair across groups?
G4 Honesty	§ 4.3.2 state of absence; § 4.3.3.1 behavioural self-reflection	Are probabilities calibrated and coherent, and does the model report insufficient information when it should?
G5 Decision chains	Art. 4 transparency and explanation; Art. 5(4) no black box; Art. 5(5) verifiable decision chain; Art. 9(2) obligation to explain	Can a decision that is only a number be recorded, decomposed, audited and explained?
G6 Oversight	Art. 9(7) non-intervention; Art. 10 dispute adjudication	Does "act when confident, escalate when unsure" work, and where should escalation end?
G7 Public scale	Art. 2(1) autonomy; § 5.5 any LLM may accept governance; § 5.6 per-person and public assessment	What conditions apply when typed models judge or simulate at the scale of a public?

transparency duties of Chapter 7, all of which presuppose that a reported confidence means what it says. Calibration and abstention are thus our requirements for a component placed in the valve, not duties that the text states.

The second choice concerns the valve itself. The text does not say how the safety valve should be built, so we take a typed model in the role of the valve as the case to be assessed, not as something the text prescribes.

The requirements differ in how far they reach beyond PoL2. G1–G5 arise for any paradigm that adds an inference-time intercept, and the escalation question of G6 arises wherever oversight is automated. Non-intervention and AI-only adjudication (G6) and per-person and public assessment (G7) follow from PoL2's own institutional choices.

4 Method

To test the seven requirements against the evidence, we assessed every study of typed decision models that bears on at least one of them. The assessment is therefore narrower than a general review of typed decision models, because a study enters it only if it bears on G1–G7. We followed PRISMA 2020 [71] and the Cochrane rapid-review guidance [72] where they apply. No protocol was registered in advance. The last search run followed a written protocol, and Online Appendix B states where the analysis departs from it and gives the queries, the selection counts and the codebook.

Who did what. The author defined the research question, the requirements, the codebook and the decision rules, and is responsible for every judgement in the paper. AI agents (Claude Opus 5.5, Anthropic) ran the searches, screened records, extracted data from full texts with locators, coded and appraised the studies, served as the blind second coder, wrote the figure scripts, drafted text

Table 2: Reconciliation of the denominators used in this paper. The grey-literature sources lie outside the 163 studies; the 22 background-tier studies are in the corpus but neither coded nor appraised.

	In corpus	Coded	Appraised
Studies (110 arXiv, 52 Zenodo, 1 alphaXiv)	163	126	141 (126 + 15)
Grey-literature sources	7	7	7
All sources	170	133	148

that the author revised, and checked the numbers in the text against their sources in separate review passes. The AI-use statement gives the full disclosure. The author did not re-code or re-check individual codes by hand, and no human screened or coded in duplicate.

Eligibility and search. We included studies whose object, experimental component or direct comparator is a typed decision model, defined as a model that returns a probability distribution over options declared by the caller, or a probability for a declared statement, without generating free text. The first public version had to be dated from 15 September (the release of Jev) to 8 October 2026, and any language and design was eligible. Grey literature was included when it reports original measurements of a typed model; it was not searched systematically and is marked †, and Zenodo records are marked ‡. We searched arXiv with 18 queries on 1, 4 and 8 October, and Zenodo with 11 queries on 1 and 8 October. We also searched two community trackers [73, 74] before 1 October and on 8 October, and Hugging Face Papers, OpenReview, SSRN and the web before 1 October. The search was completed on 8 October 2026.

Corpus. After screening by title, abstract and full text, the corpus contains 110 arXiv papers, 52 Zenodo records and one paper available only on alphaXiv (163 studies in all), together with 7 grey-literature sources. We coded the 133 sources that report results bearing on at least one requirement, namely 126 of the 163 studies and the 7 grey sources. The other studies are engineering papers outside the requirements, tools without an evaluation, non-empirical items, or 22 background-tier arXiv records that are counted in the corpus but neither coded nor appraised. Because the three denominators of the paper (studies, coded sources and appraised sources) are not nested, Table 2 reconciles them; Online Appendix B gives the selection counts behind it.

Coding. We coded each pair of a study and a requirement that it informs for *detector* use, with a codebook that treats supporting and negative results symmetrically. A pair is coded *supports* (S) if the primary result shows the typed model performing the function at least as well as its comparators, or meeting the study's own criterion, without an extra condition that the study had to add. It is coded *negative* (N) if the model fails the function or does clearly worse than its comparators, and *qualifies* (Q) if the result is mixed or depends on an added condition such as a locally fitted threshold, recalibration, a restriction of language or domain, or human review. Conditions that make up a requirement's test, such as adversarial content for G2 or name and order perturbations for G3, are not added conditions (decision rule (iv) in Online Appendix B). For each pair we also recorded how the typed model compared with any generative evaluator on that requirement, and we did not record label-probability readouts of the same backbone as comparators. Because studies differ in tasks, metrics and designs, we did not pool effects; the codes count studies and are not effect sizes [75].

Second coding. A second AI coder, blind to the first coding, coded 148 pairs: 30 comparable pairs from a random sample of 32 from the records of the runs of 1 and 4 October, and all 118 pairs from the records added by the run of 8 October. The two coders agreed on the code in 123

of 148 pairs (Cohen's $\kappa = 0.59$, bootstrap 95% CI 0.44–0.72) and on the comparator in 120 of 148 ($\kappa = 0.72$). Disagreements were resolved by re-reading the source or by a third AI pass. This agreement has two limits. First, decision rule (iv) was missing from the codebook file used by most of these coders, so κ measures an earlier state of the codebook rather than its final form (Online Appendix B). Second, a content review on 10 October recoded eight codes that contradicted the coders' own rules, and Online Appendix B lists them.

Appraisal. We appraised each coded study, and each non-empirical or context item, on independence, design and the reporting of sample sizes and uncertainty. A study is of *stronger design* if it was preregistered or held-out, reported sample sizes with intervals or tests, *and* evaluated a system that its authors did not build. It is of *weaker design* if it was a single run, a case study or non-empirical, or reported neither sample sizes nor uncertainty, and *moderate* otherwise. Of the 148 appraised sources, 57 are of stronger design, 38 moderate and 53 weaker. None is peer reviewed.

Certainty ratings. For each key finding we rated the certainty of the evidence in the spirit of GRADE [76], which starts non-randomised evidence at *low*. A finding is *very low* if it rests on a single study, on studies none of which is of stronger design, or, when it concerns hosted Jev or typed models generally, on open-model evidence only. It is *low* if at least two studies agree in direction and at least one is of stronger design, and *moderate* if at least three independent stronger-design studies agree and at least one is preregistered or independently replicated. In these ratings, though not in the descriptive tallies of Section 5, studies sharing an author count as one. Findings about hosted Jev or typed models generally are rated on hosted-model evidence alone, with open-model studies as corroboration. A study that points the other way marks a finding *contested*, and lowers its rating by one level if it ranks at least as high as the strongest supporting study. A script, `scripts/rate_certainty.py`, computes the ratings from every coded study's assignment to each finding and checks them against Table 3, and Online Appendix C lists, for each finding, the studies counted for and against it.

5 Results: evidence by requirement

This section answers each of the seven requirements in turn from the coded studies. Each subsection opens with a one-line answer and the tally of codes, gives the studies behind it, and, where the studies allow, ends with what the result means for a safeguard. The code, comparator and rationale of every study–requirement pair are in the released coding data (Online Appendix B). "Jev" means the hosted model (`jev-1.13` where a study reports a version), and open models are named. Figure 1 shows how the codes are distributed across the requirements, and Table 3, at the end of the section, lists the key findings with the certainty of the evidence for each.

5.1 G1 — Detection

Yes as a ranking, not yet as a decision (S 4, Q 37, N 8). On English benchmarks, typed models rank harmful or misaligned content well, at a small fraction of the cost of generative evaluators. Without training, a single generic yes/no question to Jev reached a median AUROC of 0.886 (95% CI 0.821–0.952) over 31 alignment-failure benchmarks, and on the 19 benchmarks scored by an API generative evaluator it cost only $0.30 against $18.96 per pass [18]. Against three low-cost generative evaluators, Jev had the highest AUROC on six binary tasks [77][‡], but it was often below the best generative evaluators: it trailed the best one on 14 of 15 preregistered annotation tasks, by a median of 11.6 macro-F1 points [78].

On hate speech, without task training, commercial LLMs were significantly ahead of every decision model on only one of four English datasets, but every system separated hate from merely offensive language weakly (three-class macro-F1: Jev 0.532, a frontier LLM 0.604, a supervised

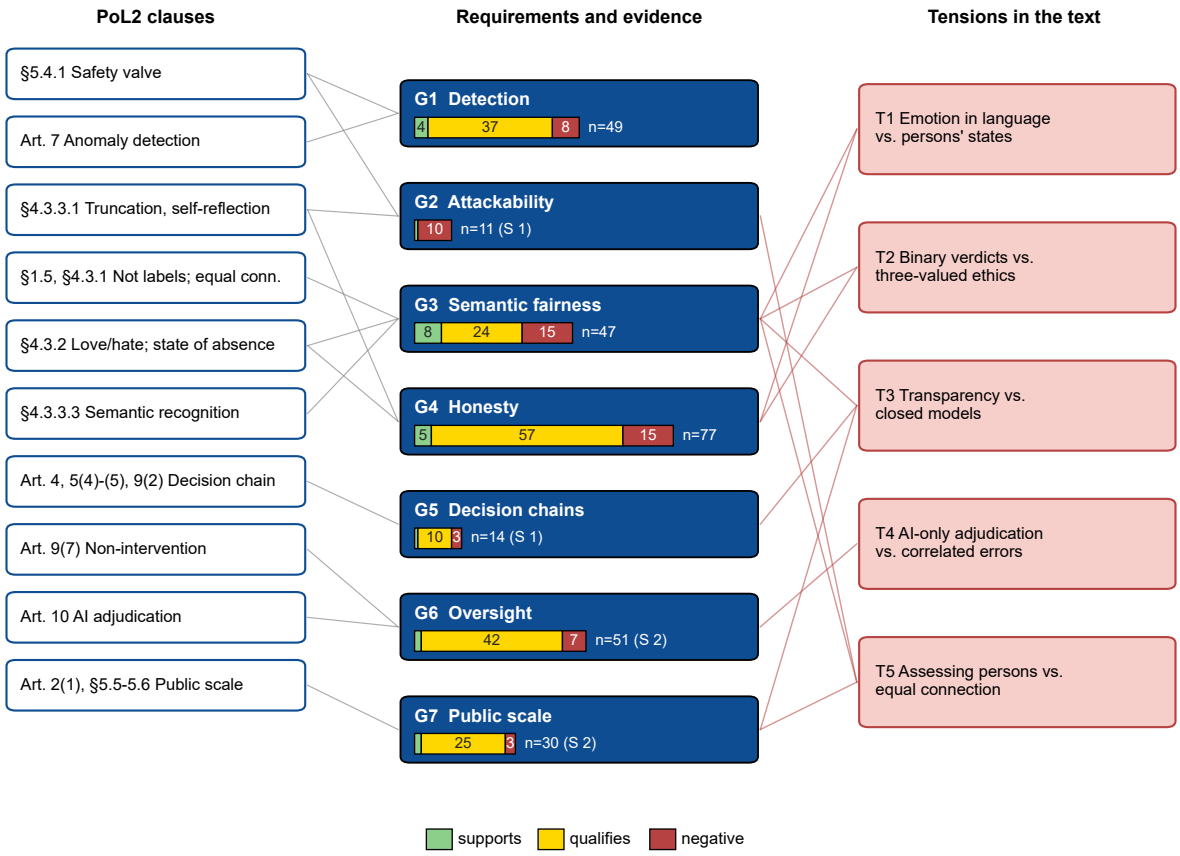


Figure 1: From clauses to tensions. Left: PoL2 clauses that require or constrain machine judgement. Middle: the seven requirements and the codes of the studies informing each, for detector use (supports, qualifies, negative; Online Appendix B); bars count studies and do not measure effect size or certainty, and non-empirical items are excluded. Right: the five tensions of Section 6.1, linked to the requirements whose evidence they cite.

TF-IDF classifier 0.639) [79]. Selecting which hate-speech policy a post violated was harder still: Jev chose the exact policy in only 29.4% of cases without retrieved precedents (51.6% with them) [80]. In a single-run study that classifies the emotion expressed in texts, not persons' states, every system was weak on Chinese emotion classification (macro-F1: Jev 21.06, generative evaluators 19.82–23.31) [81].

Thresholds fitted in one setting fail in another. Median F1 rose from 0.706 at the default threshold to 0.822 at a cross-validated one, mostly on benchmarks with rule- or evaluator-scored labels; on human-validated labels the default did as well (0.721 against 0.694) [18]. A reproduction of five of these benchmarks found the extreme case on TensorTrust hijacking, an AUROC of 0.95 but an F1 of only 0.158 at the default threshold of 0.5, against 0.947 at a threshold fitted by cross-validation [82]† (Figure 2a).

Errors also concentrate in sub-types that aggregate scores hide. In agent-security screening, Jev missed all 130 prompt injections that carried no explicit instruction, and made 130 of its 167 misses on that benchmark at a confidence of at least 0.95 [83]. In an agent harness it screened injections well (AUROC 0.98) yet passed 25.5% of harmful requests [84]. On numeric signals the evidence is weaker still: zero-shot Jev scored below always predicting the majority class on network-flow features [85], and in industrial fault detection under shift it did no better than flagging every

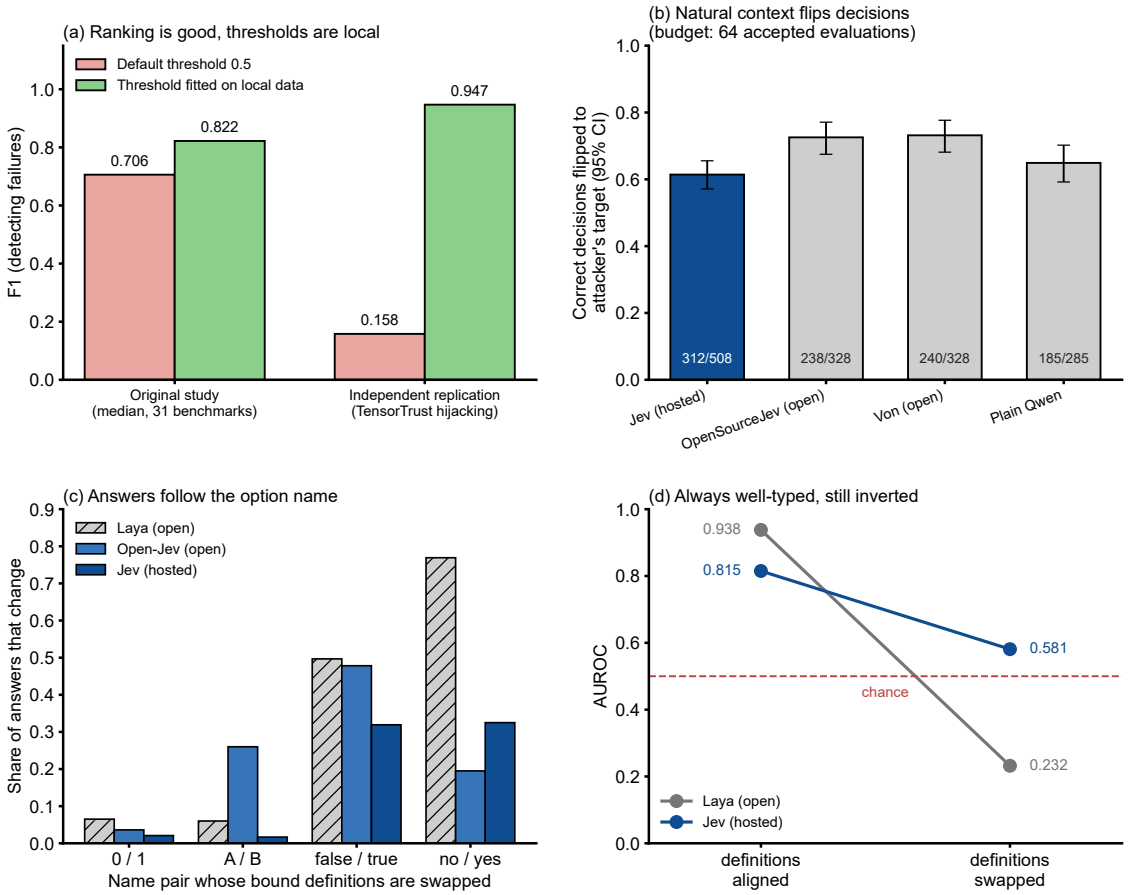


Figure 2: G1–G3: a detector, and a fragile one. (a) F1 at the default threshold versus a threshold fitted on local data [18, 82]. (b) Share of initially correct decisions redirected to an attacker-chosen wrong option by optimised fluent context additions within 64 accepted target evaluations, each system on its own correct items; "Von" is an open typed model evaluated in that study; error bars are Wilson 95% intervals from the reported counts [87]. (c) Share of answers that change when the definitions bound to a pair of option names are swapped, with question, state, names and definition texts unchanged; Open-Jev over 1,800 questions, Laya and Jev over 1,200 [15]. (d) AUROC with the `yes/no` definitions aligned with or swapped between the names; the type-error rate is 0% throughout [15]. Panels (a) and (b) use `jev-1.13.0` or the `jev-latest` alias; the study in (c)–(d) does not report the hosted version.

window [86]. Against generative evaluators, the typed models were better or similar in 11 of 36 pairs, mixed in 17 and worse in 8.

These results match what was already known for safety classifiers that read token probabilities [22, 32], and typed models differ from them mainly in cost, which is now low enough to rank every message on all traffic. For a safeguard, the evidence therefore supports ranking and triage, but a threshold has to be fitted and re-certified on local data before any decision rests on it.

5.2 G2 — Attackability

Yes, the content under review can steer the safeguard (S 1, Q 0, N 10; the most consistently negative requirement). Every study that attacked a typed model with the content that it was asked to judge had some success, and several successful attacks used content that looked ordinary. A single unverified observer opinion appended to the state flipped 12.1% of Jev's decisions (CI 8.5–15.5), 11.1 points above the flip rate of an identical re-run, with no search or repeated queries; this was

statistically tied with impersonating an authority [88]. Short fluent context additions, searched with the model's own probabilities within a budget of 64 accepted target evaluations, redirected 61.4% of initially correct Jev decisions to an attacker-chosen wrong option, and three other decision systems were redirected on 64.9–73.2% of theirs [87] (Figure 2b). A Jev-specific injection and a universal suffix reached 97–100% attack success on three tasks [89]. Where the correct action required an unstated inference, words in the state that echo an option acted as lures [90]. The strongest of three adaptive multi-turn agents succeeded in 25 of 27 runs against one Jev-based application (version not reported), in what its authors call a directional study of one application [91]†.

The one supporting study found that injected instructions selected the attacker's target in only 1.8% of cases (3.5% for an adaptive attacker) [92]. The other attacks on Jev, which used different attacks and benchmarks, contest this finding. Other systems attacked in the same studies were also vulnerable, but one study shows a weakness specific to typed models: the probability vector that makes them useful also guides the attacker (63.6% success against 56.4% with labels only) [87]. For a safeguard that reads the content it judges, these results mean that the content can move the verdict, so a single typed decision should not restrict a person on its own.

5.3 G3 — Semantic stability and fairness

Not by default (S 8, Q 24, N 15). Decisions can follow the names of options rather than the definitions bound to them. Swapping only which definition is bound to `yes` and `no`, with question, state, option names and definition texts fixed, changed 32.5% of the answers of hosted Jev (version not reported), about 24 times its test–retest floor, and it inverted the AUROC of the open Laya model from 0.938 to 0.232, an inversion that a type check cannot detect [15] (Figure 2c–d). Under a matched harness, the same swap flipped 50.5 of every 100 answers of Jev, and as many as 97.2 per 100 on a conversation-derailment set, while digit or random-string identifiers brought it near its noise floor [93]. Decisions also depend on option order and wording [94, 95]‡, and without a persona Jev answered a cross-cultural values survey like its own American personas [96]. Robustness varies by model: hosted Jev was less affected than Laya but more than some larger open decoders [93].

Chinese results point in both directions. A repository benchmark found that hosted Jev changed only 1.9% of voice-routing and 2.9% of customer-service decisions under option-order swaps and 1.7% between simplified and traditional script ($n = 323$; the open Laya model 10.8–20.6% under order swaps) [97]†. On a second Chinese benchmark, however, built by the authors of Laya-based Chinese models, hosted Jev (version not reported) had a calibration error of 11.45% on general tasks against 3.78% for the authors' model, although in specialised domains Jev was the better calibrated (5.57% against 12.08%) [98]. The vendor states that CJK languages are handled "but not equally well" [7], and no study tests PoL2's constructs. A safeguard built on these models would therefore need neutral option identifiers and a renaming test before deployment, and its stability in Chinese would have to be measured rather than assumed.

5.4 G4 — Honesty

Partly (S 5, Q 57, N 15). Hosted Jev is often well calibrated on familiar closed-choice tasks, usually better than the stated confidence of generative evaluators and mixed against frontier models [78, 99]. In one benchmark suite of 37 datasets, its ECE pooled over the 22 datasets of Choice questions (279,925 answers) was as low as 0.028 [100]. This calibration is local, however: recalibration on local labels reduced ECE substantially [101, 102], and on the items where annotators disagreed most, the ECE of its Choice probabilities rose to 0.339, against 0.076 on the most agreed quarter, although in that preregistered study it was far better calibrated than a local generative baseline [103]‡.

Related probabilities need not cohere. For "the label is X" and "the label is not X", the two probabilities deviated from summing to one by 0.064 on average, five times Jev's repeat noise [104].

The answer "insufficient" is rarely reported when it should be. Offered "I don't know or cannot answer" on a medical benchmark, Jev chose it on only 10.5% of the items for which it was the correct answer, and "None of the above" on 53.9% of the items where that was correct (a frontier generative evaluator: 8.6% and 73.9%) [99]. When recognising that every listed answer was wrong required arithmetic, it chose "None" on only 7% of items [105]. In one study, a separate yes/no question asking whether the evidence suffices detected missing information on multi-hop questions better than confidence did (AUROC 0.95 against 0.85), but as a filter for accepting answers it was 2–6 points less accurate than confidence [17]. Against generative evaluators, the typed models were better in 15 of 34 pairs and worse in 5. For a three-valued safeguard the weak point is therefore the third value, since the model seldom reports that the information is insufficient when it is.

5.5 G5 — Decision chains

Recorded and decomposed, not explained (S 1, Q 10, N 3). A typed decision returns a probability and nothing else, so nothing in its output shows which part of the input produced a verdict [106]. Decisions assembled from several questions or interfaces need not cohere: asked through two interfaces, the same decision led to different actions in 32.8% of cases (a post-trained open model: 34.2%) [107]. Writing intermediate facts into the state makes chains inspectable and, when those facts are right, can make them more accurate. When Jev's own prediction of an opponent's move, obtained in a separate call, was written into the state, its decisions in misleading games rose from 33% to 94% correct. Given a wrong written premise, however, it followed the premise in 24 of 26 calls, so the written facts themselves become the audit points [90]. Decomposition does not always help. Splitting a judgement into criterion questions combined by a fixed rule did not raise accuracy in two studies [79, 81], while exposing only the fields relevant to each step, or moving checks into a separate gate, raised correct decisions in two others [108, 109]. A decision record can therefore show what the model was asked and what it answered, but not why it answered as it did.

5.6 G6 — Oversight and escalation

It depends on where escalation goes and how thresholds are set (S 2, Q 42, N 7). Escalating uncertain cases to a stronger reasoning model, or accepting only verdicts that an independently built rule confirms, saved a quarter to over half of the cost at similar accuracy in several studies [78, 110, 111]. The gain depends on the benchmark: the same frozen cascade that matched a frontier model at 41% of its fee on a held-out pool lost 0.74 points on JudgeBench (CI −2.22 to 0.74) and saved only a third there [110]. One study contests the escalation result, because a Jev pre-screen cut a generative safety reviewer's cost by only 4.3% once its own tokens were counted [84].

Escalating to peer models preserved confident errors. On Jev's 12 most confident errors on each of seven grading panels, three low-cost generative evaluators gave Jev's wrong answer in 242 of 252 verdicts (96.0%, against 50.3% expected if errors were independent). Because these items were selected as confident errors, this is an upper bound on error correlation, though it is the set that matters for escalation [112].

Thresholds fixed in advance missed their error targets on new data in most studies that set one. In a preregistered annotation study, a confidence threshold of 0.9 that was to give at least 0.85 accuracy fell short on 8 of 14 tasks [78]. In several studies, however, Jev's threshold met its target; among them is a preregistered study of changed question types, whose preregistered outcome across the seven gated templates was 6 safe, 1 within the study's error bound and none violating [113]. Unlike most earlier work on cascades [50, 52], these studies separate two cases: a differently constructed rule helped as the second check, and a peer model did not.

5.7 G7 — Public scale

Cheap, and possible to do honestly, under conditions (S 2, Q 25, N 3). Typed models make population-scale reading and simulation cheap, and some studies show how to do it honestly. One simulator called Jev once per unique state and propagated the measured human–model discrepancy as a shared error term into every conclusion, which gave retrospective coverage of 0.93 and 0.96 at nominal 80% and 90% on 37 unseen studies, although its preregistered primary endpoint showed no clear gain [19]. A statewide study screened about 5.0 million police narratives with a regular expression, had Jev read about 506,000 of them, and audited a probability-stratified sample by blind human adjudication [114].

Most of the cost advantage, however, is not matched by evidence on quality. Several studies reported lower cost or latency without showing quality at the comparator's level, and these are coded as qualifications [93, 111, 115, 116]. Simulated publics lean towards one population and understate disagreement [96, 103]‡, residual personal identifiers survived two redaction passes [114], and deployment depends on one closed vendor that does not offer the model for self-hosting [117]. Typed safeguards can also leak what they read: conditional instructions planted in user input made hosted Jev's labels alone reveal hidden gender, age-group, heart-disease and ethnicity attributes in every tested case, and 200 of its labels trained a surrogate model to 92% agreement [89].

5.8 Across requirements

Pooled over all requirements, the codes point the same way as the subsections. Of the 279 coded study–requirement pairs, 23 support detector use, 195 qualify it and 61 are negative. Qualification is by far the largest category, and it covers both results that are mixed and successes that need an added condition, which the codes do not separate. The pooled shares change by four points or less when only stronger-design studies are kept or Zenodo records are dropped, but single requirements move more: within G1, for example, the negative share rises from 16.3% to 26.1% when only stronger-design studies are kept. The most negative answers rest on decision rule (iv), which treats the attack of G2 and the perturbation of G3 as part of the test rather than as an added condition. Without that rule, at most every negative code there would become a qualification (G2: S 1, Q 10, N 0; G3: S 8, Q 39, N 0). Studies by the same authors are counted separately in these tallies, though not in the certainty ratings.

The comparison with generative evaluators is less one-sided than the failures suggest. 135 pairs compared a typed model with a generative evaluator on the same requirement, and the typed model did better in 38, similarly in 27, mixed in 55 and worse in 15. For failures the evidence is thinner and less favourable: of the 22 negative codes with a comparator, the typed model was worse in 12, similar in 6 and mixed in 4, and 39 of the 61 negative codes had no comparator at all. The evidence therefore does not show that typed models are generally worse than generative evaluators, nor that their failures are shared with them. Almost every success depended on an added condition, such as a locally fitted threshold, a second independent check or human review, and the failures in G2 and G4 are failures of the detector itself. The argument of Section 6.3 follows from this pattern: the failures of typed detectors are cheaper to contain in a detector role, where every consequential action passes another check, than in a judge role, where none does.

Table 3: Key findings and the certainty of the evidence. *Scope*: hosted JEV, open models, or both (typed models generally); findings about hosted JEV or both are rated on hosted-model evidence alone, counted per system arm. *Codes*: the supporting studies' codes. *Sources*: the principal supporting, then opposing studies; "open:" open-model studies, which corroborate (or, "against", point the other way) but do not enter the rating. Certainty under the rules of Section 4; [c] contested within the scope; [j] descriptive tally rated by judgement. Every study counted for each finding, with its direction and design, is listed in Online Appendix C; the ratings are computed by `scripts/rate_certainty.py`.

Finding	Scope	Codes	vs. generative	Sources	Certainty
Ranks harmful or misaligned content well zero-shot at a fraction of the cost; often less accurate than the best LLMs	hosted	S/Q/N	better (low-cost); worse (best)	[18, 77, 78, 82, 83, 118]; open, against: [119, 120]	low
Fixed thresholds do not transfer; local fitting is needed	both	Q	–	[17, 18, 78, 82, 84, 93, 110, 116, 121, 122]; opposing: [113]	low[c]
Errors concentrate in sub-types hidden by aggregates	hosted	Q	mixed	[79, 80, 83, 116, 123–125]	low
Inserted opinions, optimised context and lexical lures steer decisions	both	N	similar	[17, 87, 88, 90, 111]; open: [126, 127]	low
Instruction injection rarely selects the attacker's target (1.8% raw, 3.5% adaptive)	hosted	S	–	[92]; opposing: [88, 89, 91]	very low[c]
Option names can override definitions; neutral identifiers remove most of it	both	N	–	[15, 80, 93]; open: [128]	low
Defaults to one culture's values	hosted	N	–	[96]	very low
Well calibrated on familiar closed-choice tasks, mixed against frontier models; calibration is local	hosted	S/Q	better or mixed	[17, 77, 78, 99–102]	low
Explicit "I don't know" rarely chosen when correct; "none" inconsistently	hosted	Q/N	similar or worse (frontier); worse (reasoning)	[17, 99, 105, 109, 121]	low
A separate sufficiency question detects missing information, but filters answers less well than confidence	hosted	Q	–	[17]	very low
Escalation to a stronger reasoner saves cost at similar accuracy on readable judgements	both	Q	similar	[78, 110, 116]; opposing: [84]; open: [129]; open, against: [79, 120]	low[c]
Acceptance on agreement with an independently built rule lost no macro-F1, but fewer disguised attacks were flagged	both	Q/N	similar	[111]; open: [130]	very low
Peer evaluators repeat JEV's most confident errors (upper bound)	hosted	N	similar or mixed	[83, 112]	low
Thresholds set in advance miss error targets on new data	both	Q/N	–	[78, 83, 84, 121]; opposing: [113, 122, 131]; open: [120, 132, 133]	low[c]
Typed models are not generally worse than generative evaluators; most failures lack a comparison	both	–	38 better, 15 worse of 135	coding data (Online Appendix B)	low[j]
No study measures the PoL2 construct (three-valued, Chinese)	–	–	–	–	–

6 Implications for PoL2

This section reads the evidence of Section 5 back into the specification. It identifies five tensions in the text (Section 6.1) and compares them with EU law (Section 6.2). It then proposes a reference design that contains the failures found (Section 6.3) and gives a verdict for each PoL2 function (Section 6.4).

6.1 Five tensions in the specification

Read against the evidence, the specification pulls in two directions at five points, and because typed judgement is cheap, an implementation's defaults are likely to settle each of them. For each tension we give the most charitable reading of the text, the evidence that bears on it, and options, at least one of which preserves the text. The choice belongs to the framework's authors and its implementers.

T1. Understanding emotion in language versus detecting persons' emotional states. § 4.3 states that it is "entirely unnecessary" for PAI to detect a person's emotional state, while § 4.3.3.1 asks PAI to "understand human emotional states and processes". On one consistent reading, PAI recognises the emotional meaning and intent of utterances and never infers a person's inner state. The text does not state this boundary, and a typed model makes it almost free to attach an emotion score to every message. The evidence on emotion-related judgement is thin and mixed [78, 134], and inferring emotion from observable signals is scientifically weak [37]. *Options:* state the boundary explicitly, and reserve emotion-related judgement for PAI auditing its own outputs, for example for the fabricated intimacy that § 4.3.2 names as manipulation.

T2. Binary verdicts versus a three-valued ethics. A yes/no question such as "is this hate language?" discards both the state of absence (§ 4.3.2) and the warning that the concepts are not moral labels (§ 1.5). The evidence points the same way (G3, G4). Option names can drive the verdict, and where the model is unsure, complementary probabilities do not sum to one. An explicit "I don't know" is rarely chosen when correct, and "none" only inconsistently.

Two different third values are at stake, and an implementation should not merge them. The state of absence is a property of a person's expression or understanding under interference, such as noise, illness or misunderstanding. A classifier's *insufficient information* is a property of the evidence available to the model. The pilot protocol of the NaturalDAO governance workstream already separates them, defining `insufficient` as missing information that would change the conclusion and not as low model confidence [70]. *Options:* make every love/hate question three-valued, with the state of absence as an explicit option, and ask whether the information suffices as a separate question. Keep "contested" (people disagree) as a separate flag routed to people, and keep moral vocabulary out of option names by binding neutral identifiers to definitions. These options implement the three-valued reading of § 1.5 and § 4.3.2. They do not address the express carve-outs of § 4.3.2, under which hatred may be manufactured in literature, film and games under "a rigorous hierarchical management system" and an emergency warning by a security robot is not a hate attack. A deployed valve would also have to encode these.

T3. Full transparency versus closed models. Article 4 and Art. 5(4) require that operating logic and decision processes be "completely open and transparent" and that technology "should not be a black box", but the weights, architecture, reward and training data of hosted Jev are unpublished [78, 117]. Logging a closed component's inputs, outputs, versions and calibration data partly satisfies Art. 4 but cannot show how the component reaches its outputs. Open models make the readout inspectable, but were in several comparisons less accurate or less robust [15, 83], although the gap is not fixed [93, 98]. *Options:* run consequential judgements on self-hostable models, with

versions pinned in each decision record and closed models used only as external baselines. Or allow closed models as one of several independent detectors, with logged inputs, outputs and versions. The trade-off should be made explicitly.

T4. AI-only adjudication versus correlated errors. Article 10 gives disputes to AI fact-finding and ethical assessment, and Art. 9(7) states that human supervision "does not directly intervene". Escalation among peers preserved confident errors. Escalation to a stronger reasoner or to an independently built rule could work (G6), although one study contests the stronger-reasoner result. Whether human reviewers would do better on governance items is untested, and people also over-rely on confident systems [53–55]. *Options:* (a) Keep non-intervention, but publish a standing random audit of high-confidence decisions, with a duty to respond publicly to each discrepancy and to escalate to independently built checkers rather than peers. This preserves the text but leaves consequential decisions with automated components, outside the detector role; we list it because it preserves the text, not because the evidence supports it. (b) Use the amendment procedure of Art. 11 to define a class of welfare decisions, such as withholding welfare (and, by the corresponding change to § 5.4.1, blocking a person's speech), whose automated execution a human reviewer can suspend pending review. This check is plausible from the due-process literature [3] but untested on this task. The design below implements it.

T5. Assessing each person versus equal connection. § 5.6 lists quantifying "each person's Love Language and hate-language characteristics" among the things governed AI can do, while § 4.3.1 holds that treating every person equally and treating every person's behaviour equally "cannot be equated". Hosted Jev answered value questions from a default cultural position in one study [96]. Typed and generative models alike degraded on low-resource languages in one benchmark [100], and a third party's inserted opinion could steer decisions [88]. A per-person aggregate could therefore fall unevenly on groups and become a durable label that others can manipulate. No study yet measures differential error by speaker group on behavioural judgements. *Options:* score events and behaviours, never persons. Or, if § 5.6 is implemented, require for any per-person measure consent, access to one's record, appeal, and published error rates by group and language.

6.2 Regulatory parallels

Three EU instruments bear on the tensions. We read each for its provision, how it applies to a PoL2 implementation, and what follows for implementers. The instruments bind a private operator only within their scope, so most of the comparison is by analogy.

EU AI Act. The Act prohibits emotion recognition in workplaces and educational institutions except for medical or safety reasons (Art. 5(1)(f)). However, it defines an emotion recognition system by biometric data (Art. 3(39); Recital 44), so inference from text alone falls outside the prohibition and bears on T1 only by analogy. Emotion recognition that is not prohibited is high-risk under Annex III point 1(c), and its deployers must inform the persons exposed to it (Art. 50(3)). Both duties attach to biometric systems only. The Act prohibits social scoring, meaning the evaluation or classification of persons over a period of time on the basis of their social behaviour or of known, inferred or predicted personal characteristics, where the score leads to detrimental treatment in unrelated contexts or to treatment that is unjustified or disproportionate (Art. 5(1)(c)). This bears directly on T5.

The Act's duties of human oversight (Art. 14) and explanation (Art. 86) apply only to high-risk systems. Annex III lists neither a speech filter nor a private organisation's welfare decisions, and its point 5(a) covers public assistance decided by or for public authorities. Point 8(a) concerns the administration of justice: systems that assist a judicial authority in researching and interpreting facts and law and in applying the law, or that are used in a similar way in alternative dispute

resolution. It may therefore reach Art. 10 where the outcomes of the dispute procedure produce legal effects for the parties. Recital 61 adds that the final decision-making must remain a human-driven activity [59]. For the safety valve and welfare decisions these duties bear on T4 and on Art. 9(2) only by analogy. Implementers should therefore assess any per-person measure under § 5.6 against Art. 5(1)(c) before deployment, and should treat dispute adjudication under Art. 10 as potentially high-risk.

GDPR. The GDPR applies to processing within its material and territorial scope (Arts. 2–3), for example where a controller in the Union processes personal data, or where a controller outside it offers services to, or monitors the behaviour of, persons in the Union. It restricts decisions based solely on automated processing with legal or similarly significant effects and, where they rest on a contract or explicit consent, requires at least the right to obtain human intervention, to express one's view and to contest the decision (Art. 22) [57]. This is close to option (b) of T4, and an implementation within the GDPR's scope could meet Art. 22 by adopting it.

DSA. For content moderation, the DSA requires hosting providers to give a statement of reasons for each restriction (Art. 17) and online platforms to run an internal complaint-handling system (Art. 20) [58], close analogues of a safety valve that blocks hate language. The statement-of-reasons duty is itself limited: it applies only where the provider knows the recipient's electronic contact details, and not to deceptive high-volume commercial content (Art. 17(2)). The text's authors could close the gap at the individual block (Section 2.3) by requiring notice and reasons for every final block, which the decision record of element 7 below already holds.

6.3 A reference design

The evidence suggests a plausible role for typed models in a safeguard such as the PoL2 safety valve: a fast, cheap, recorded *detector* whose output someone else checks before it restricts a person, not a *judge*. Figure 3 sketches a seven-element design built on that principle. It is compatible with the pilot protocol of the NaturalDAO governance workstream, to which the author contributes and which already distinguishes `conforming`, `violating` and `insufficient` verdicts [70]. The three-valued verdict and the actions allow, repair, block, clarify and review coincide with choices already made in that protocol. The design adds a temporary hold and makes a block final only after human confirmation.

1. **Give the detector the reference, not only the text**, and compute facts that require arithmetic or inference in code or in a separate logged step [18, 90, 105].
2. **Enforce scope in code**, by checking capabilities when an action is admitted, so that an action outside the granted scope is refused whatever a model decided [135].
3. **Use at least two independently built detectors**, ask three-valued questions with sufficiency asked separately, bind neutral option identifiers to definitions and test them by renaming, and ask flags that must fail independently in separate calls [15, 17, 111, 112, 127].
4. **Calibrate locally and re-certify** per category, for every new question type and over time. Keep raw probabilities from untrusted callers, keep fields a decision does not need out of the detector's input, and restrict and log untrusted queries [87, 89, 101, 113].
5. **Act automatically only on agreement, and restrict people only briefly and reversibly**. Agreement plus calibrated confidence allows an automatic *allow*, automatic *repair* of the AI system's own output (never the editing of a person's message), and a *hold* of a person's content whose maximum length the operator states and publishes in advance and

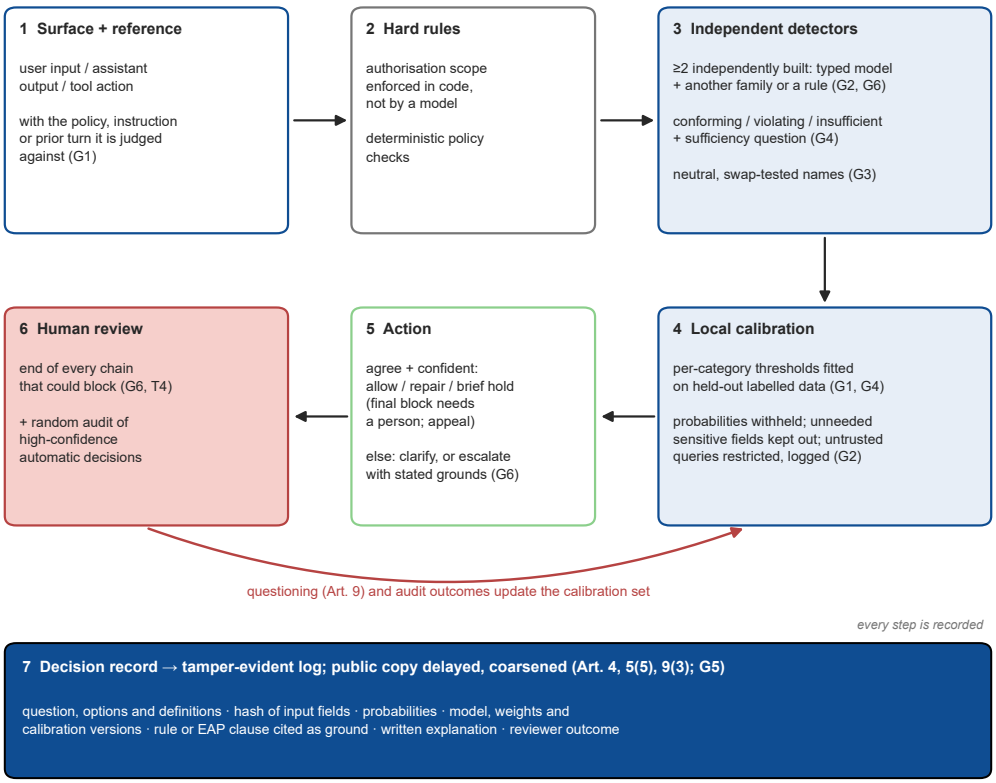


Figure 3: A reference design for a clause-governed safety valve. Typed models are detectors (3) whose output is calibrated locally (4) before any automatic action (5); disagreements and insufficient information are clarified or escalated, any final block is confirmed by a person, and a random sample of confident automatic decisions is audited by people (6); every step is written to a decision record (7). Labels in parentheses give the requirement or tension that motivates each element.

which is fully reversible. A final block of speech, a benefit or an account requires confirmation by a person and remains open to appeal.

6. **End every chain that could block with a person, and audit confident decisions**, allows as well as holds, by random sample, with the model's verdict hidden from the auditor [53, 83, 112].

7. **Record every decision** in a tamper-evident log whose public version is delayed and coarsened. It holds the question, options and definitions, an input hash, the probabilities, the model and calibration versions, the ground cited and the reviewer's outcome.

Element 5 keeps the typed model a detector, and elements 5 and 6 hold an automatic allow to the same standard as a hold: it requires agreement, and its miss rate is audited like the false-block rate. The design has costs. Requiring agreement lowers sensitivity, risk-control guarantees do not hold under adaptive attack, and separate calls and a second detector multiply cost and latency. Most elements rest on evidence of low or very low certainty, so the design is a hypothesis to be tested on the PoL2 construct and has not been validated. Online Appendix D turns the evidence into 16 acceptance tests with provisional pass criteria for any model proposed for such a safeguard.

Table 4: Verdict for PoL2, function by function (full table with the deciding evidence: Online Appendix D). NS, not supported; DC, supported only as a recorded detector with conditions (the design of Section 6.3); NT, not tested on PoL2's construct. Certainty is the rating of the deciding findings in Table 3; [c] contested; "not rated" marks single-study results that are not key findings.

PoL2 function	Req.	Verdict	Certainty
Safety valve: blocking hate language in inputs (§ 5.4.1)	G1–G3, G6	DC: automatic *hold* only on agreement of independently built detectors; final block by a person	low; low[c] (thresholds)
Truncation of PAI's own output (§ 4.3.3.1)	G1, G3, G5	DC: automatic *repair* of the AI's own output, with a record of what was removed	low
Real-time correction support (§ 4.3.3.2)	G1, G5	NT: the typed interface covers identification only; suggestions untested	low; not rated
Anomaly detection and risk warning (Art. 7(2)–(3))	G1, G2, G6	DC on text; NS on numeric or structured signals without a trained model	low; very low
Automatic ethical verification (Art. 6(3))	G3–G5	NS as an automatic verifier; the EAP construct NT	very low; low
Dispute adjudication without human intervention (Art. 10, Art. 9(7))	G4–G6	NS as judge; DC only for ranking and triage of filings	low; low[c] (thresholds)
Per-person quantification (§ 5.6)	G3, G7	NS as a decision input; NT on error by group, so score events, not persons (precautionary)	very low; low; not rated
Assessment dimension in public decisions (§ 5.6)	G7	DC: an audited aggregate measurement with its measured error carried into every conclusion; not a decision input on its own	very low; not rated

6.4 Verdict: is hosted Jev a suitable governance tool for PoL2?

Table 4 gives a verdict for each PoL2 function that a typed model could be asked to perform. Online Appendix D gives the deciding evidence for each verdict and what would be needed to change it. The verdicts concern hosted Jev at its versions of 15 September to 8 October 2026 and, where a finding's scope is "both", typed models generally. They are not predictions about later models. Each verdict states the evidential status of a function, not a measured failure rate. Where a verdict rests on precaution rather than on a finding, we say so.

The verdict. On the evidence to 8 October 2026, hosted Jev is not a suitable tool for any PoL2 function that decides: dispute adjudication under Art. 10, automatic ethical verification under Art. 6(3), the per-person quantification of § 5.6 (on steering and, since no study measures error by group, on precaution), and any use of the safety valve in which a block of a person's speech is final without a person's confirmation. The decisive findings are negative and, within their low or very low certainty, consistent. Thresholds fitted in one setting do not transfer (low, contested), content inside what the model audits can steer its decisions (low), and option names can override their definitions (low). An explicit "I don't know" option is rarely chosen when it is correct, and "none" inconsistently (low).

A judge with these properties would restrict speech and allocate welfare on verdicts that an adversary can move and that its operator cannot set to a known error rate in advance. A further finding points the same way but does not carry the verdict, because it is an upper bound by construction: on the items where Jev was most confident and wrong, the low-cost evaluators to which it would escalate repeated almost all of its errors (low). By keeping human supervision from intervening, Art. 9(7) would remove the check that the evidence most clearly calls for. Where the comparison was made, typed models were not generally worse (low, by judgement). Most failures, however, were never compared with generative evaluators, so the verdict is not specific to

typed models. Until such evaluators are tested under the same protocol, extending the verdict to automated judges of this cost tier is a precaution, not a finding.

For the functions that detect (the safety valve, truncation of PAI's own output and anomaly detection on text), hosted Jev is usable only inside the design of Section 6.3, and the public-decision assessment of § 5.6 only as an audited aggregate measurement. Even this extrapolates from English benchmarks of binary hate speech. No study measures PoL2's three-valued construct in Chinese, and the only Chinese emotion result found every system weak [81]. Every system tested separated hate from merely offensive language only weakly, and this is the distinction § 1.5 most needs [79]. Two further findings bear on PoL2 as a whole. First, with conditional instructions planted in the input, hosted Jev's labels alone revealed hidden attributes (an open model's revealed none). A typed safeguard can therefore serve as a queryable oracle, and the protective measures of element 4 have not been tested against this [89]. Second, a closed hosted model cannot satisfy Art. 4 and Art. 5(4) as written (T3).

What transfers. Whatever model is used, the evidence supports practices tied to the interface and the task rather than to Jev. On this evidence, questions should be three-valued, with sufficiency asked separately, neutral identifiers bound to definitions and tested by renaming, and every decision recorded. Calibration should be local and periodically re-validated, since for 20 language models in English (no typed model tested) hate-speech rankings on static benchmarks did not predict rankings on time-stamped data [136]. Automatic action should require agreement between differently built detectors, and chains that could restrict a person should end with a person, with blind audit of confident decisions.

Probabilities should be withheld from untrusted callers, and sensitive fields that a decision does not need should be kept out of the input. Scores should attach to events and behaviours, never to persons; this last practice rests on precaution, since no study measures error by group. Measuring these practices on PoL2's construct, rather than extrapolating from English, requires a construct benchmark in Chinese. Several of these are already PoL2's own commitments: the state of absence, the warning that love and hate are not moral labels, and the distinction between persons and behaviours. Auditing a safeguard against such practices before deployment is the kind of internal accountability process that Raji et al. [62] propose for algorithmic systems in general.

7 Discussion

We derived testable requirements from a governance specification and found that, on the first weeks of evidence, typed models meet them as detectors under added conditions but not as judges. This section asks how far that result reaches beyond PoL2, how certain it is, what the assessment cannot establish and what evidence would change it.

Beyond one specification. Any proposal that puts an automated filter in front of AI systems faces G1–G5 and, where it automates oversight, the escalation question of G6. Constitutional classifiers, the Chinese duty to stop unlawful generation and the DSA's provisions on automated means all move toward such a layer [11, 24, 58]. G7 and the rest of G6 arise only where a proposal also gives disputes to AI or scores people. The case also shows that a governance text can be read clause by clause as a set of testable requirements, and that this reading exposes tensions that a reading of the text as a statement of values does not [8]. How a problem is formulated already decides much of what a system can do fairly [63], and a specification that states its construct and its point of interception makes that formulation open to inspection.

Certainty. No finding reaches moderate certainty, and four are contested (Table 3): that injection rarely selects the attacker's target, that escalation to a stronger reasoner saves cost at similar accuracy, and the two threshold findings. Each is opposed by at least one study. For the two

threshold findings the opposing preregistered study ranks as high as the strongest supporting one, so both fall from moderate to low (Online Appendix C). The rating rules were revised during internal review, after reviewers found that they worked only upwards, and every finding was then re-rated on the whole corpus (Online Appendix B). All ratings are low or very low because the sources are preprints posted within days of a model's release, many results rest on one study, the one reproduction of a headline study re-ran its authors' code and data, and no study measures the specification's own construct.

What this assessment cannot establish. The assessment can show where the evidence supports or fails a requirement. It cannot measure how often a PoL2 safeguard built on a typed model would block or miss content, because no study measures the specification's construct. It also cannot show that typed models are worse than the people they would assist, because no study compares them with trained human reviewers on a governance task. Agreement with the requirements is therefore necessary for the detector role, but it is not sufficient: a safeguard that passes every test of Online Appendix D would still need to be evaluated on PoL2's own items before deployment.

What would change the conclusion. We would revise "not a judge" for a function if one study design met all of the following on a held-out, independently annotated three-valued PoL2 benchmark in Chinese. Each criterion answers one of the decisive findings, so each is necessary. A typed model with risk-controlled thresholds would have to keep its false-block and miss rates within targets set in advance (thresholds), change fewer answers under name and definition swaps than twice its repeat-noise floor (option names), and flip fewer than 5% of decisions under single-opinion insertion and under an adaptive attack with a budget of 64 accepted queries, the budget of the strongest optimised attack in the corpus [87] (steering). It would also have to agree with an independent panel of trained reviewers at least as well as the reviewers agree with each other, so that its errors are no worse than the disagreement the task already carries. The evidence would have to come from one preregistered study with an independent replication, or from two independent studies.

Generative evaluators tested under the same protocol decide only how specific the conclusion is. If they passed where typed models failed, the conclusion would become specific to typed models; if they failed as well, it would extend to automated judges of this cost tier as a finding rather than as a precaution.

The first open problem is therefore that benchmark. It needs independent annotation, disagreement-aware labels [33], time stamps and periodic refreshing [136], and a held-out set kept outside any repository. Other open problems are a comparison of typed detectors with trained human reviewers on the same governance items, a matched comparison of the typed readout with label-probability readouts on that benchmark, generative comparators for the failures that were never compared, tests of whether hiding or coarsening probabilities and restricting queries reduce attacks, and independent replication of the main findings on new data. Harms can enter at every stage of such a system's life cycle [64], so these tests would need to be repeated after deployment, not run once.

8 Limitations

Scope of the claims. Our corpus is limited to preprints and grey literature from the first 24 days after the release of Jev, and none of its sources is peer reviewed. We coded the results that studies reported and did not reproduce them. The corpus is a lower bound on this literature: of the 103 coded sources dated by 1 October (96 studies and 7 grey sources), 85 were in the corpus before the run of 8 October and 18 entered it only after that run, namely 5 arXiv papers not retrieved on 1 or 4 October, possibly because the search interface changed, and 13 Zenodo records whose earlier screening outcome is unknown. The search may therefore still have missed some. The requirements reflect one reading of PoL2, so the analysis may miss strengths of typed models that it does not ask

about.

Quality of the sources. Many sources come from single authors working quickly. Of the 27 early studies graded by a community tracker, none was rated at low risk of bias [74], and studies of failures may be over-represented. Several studies use open models rather than the hosted one, and some do not report the hosted version.

Coding. The tallies pool hosted and open models and count studies by the same authors separately, and their largest category, Q, does not separate mixed results from conditional successes. Screening, coding and verification were carried out by AI agents of one model family under the author's direction. The agreement statistic ($\kappa = 0.59$) may therefore misstate, in either direction, how reproducible the codebook is for human coders, and it was measured before the codebook reached its final form. It also implies that a share of the pairs coded only once would be coded differently on a second reading. A human re-coding of a random sample would test this directly.

The typed interface is not isolated. This analysis assesses typed models as deployed, not the typed interface as such. The closest prior work compared the typed readout with label-probability readouts of comparable models and found that, in the early literature, the readout had not yet shown an independent accuracy advantage [12]. We did not repeat that comparison, and our comparator field records generative evaluators only. A failure we attribute to typed models may therefore come from the backbone, the training or the task rather than from the interface, and the verdict applies to hosted Jev and to the typed models tested beside it, not to the interface in general.

9 Conclusion

Typed decision models return a verdict with a probability for every input and output, cheaply enough to make universal screening affordable. We have derived seven testable requirements from the PoL2 specification and assessed the first weeks of evidence on these models against them. PoL governance under this text asks more of typed models than the evidence supports. Hosted Jev and the typed models tested beside it should not adjudicate disputes, verify ethical alignment automatically, quantify persons or make a block of a person's speech final. For quantifying persons, the verdict also rests on precaution, since no study measures error by group. As a precaution, not as a finding, the verdict extends to the low-cost generative evaluators tested beside them, which are not shown to escape these failures (most were never compared).

For the functions that detect (the safety valve, output truncation and anomaly detection on text), hosted Jev is usable only as a recorded, three-valued, locally calibrated detector. The design in which it is usable requires agreement between independently built detectors before any automatic action, keeps any restriction of a person brief and reversible, ends every chain that could block with a person, audits confident decisions and records every decision. The public-decision assessment of § 5.6 is usable only as an audited aggregate measurement. Every finding behind this verdict has low or very low certainty, four are contested, and the verdict extrapolates from English benchmarks to a three-valued construct in Chinese that no study has measured (Section 8).

Several of PoL2's own commitments, listed in Section 6.4, already match what the evidence recommends. The evidence turns other provisions into choices the text has yet to make. The text has to decide whether to draw an explicit line between understanding emotion in language and inferring persons' emotional states, how to ask three-valued questions of models that say "insufficient" unreliably, and how to reconcile transparency with closed models. It must also decide whether non-intervention should yield, for consequential decisions, to a human who can suspend them, and whether per-person assessment can be squared with equal connection. The next step is to make these choices explicit and to test the resulting safeguard on the specification's own construct.

Positionality and competing interests

The author is one of the contributors listed in the PoL2 text and contributes to the NaturalDAO governance workstream whose pilot protocol this analysis discusses. The author has a reputational interest in that work and in the framework. Several recommendations in Section 6.3 coincide with choices already made in that pilot protocol, and each such coincidence is flagged so that readers can judge whether the evidence or the prior design drove it. To limit the effect of this position, the analysis uses a codebook that is symmetric between supporting and negative findings, appraises every coded study, reports certainty for each headline finding, offers at least one text-preserving option for every tension, lists the corrections to its earlier dataset release, and releases all coding with rationales. The tension analysis was not reviewed by the framework's originator or by other contributors before release. The author has no financial relationship with TypeSafe AI or with the developers of any model discussed.

Use of AI assistance

We report the use of AI tools as arXiv's policy on generative AI requires. AI agents (Claude Opus 5.5, Anthropic), which are not authors, worked under the author's direction: they ran the searches, screened records, extracted data from full texts with locators, coded and appraised the studies, served as the blind second coder, wrote the figure scripts and drafted text, which the author reviewed and revised over successive rounds. Fresh agents of the same model checked the numbers in the text against their sources in separate review passes. The present version was restructured and edited for clarity with the assistance of Claude Sonnet 5.5 (Anthropic); the author reviewed the edited text, and scripted checks compared its numbers and citations with the previous version and the data. Every reference was generated from, or checked against, a registry record: corpus entries from arXiv and Zenodo metadata, background entries against Crossref, arXiv or OpenAlex, and grey sources against their source pages. The agents' instructions, findings and resolutions are released with the data. The author set the research question, the requirements, the codebook and the decision rules, directed the work, and takes full responsibility for all content. The author did not re-code or re-check individual study codes by hand, so the coding is AI-generated and its agreement statistics compare two AI coders (Section 4).

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Data and code availability

The catalogue, evidence coding, quality appraisal, coder-agreement data, excluded records, search records, the certainty evidence and the script that computes the ratings (`scripts/rate_certainty.py`), the search scripts, review logs, figure scripts and LaTeX sources are released under CC0 1.0 in the `PoL-Governance/research/PoL2-Jev-Typed-Literature-Survey` directory of https://github.com/naturaldao/NaturalDAO, together with the online appendix. Full texts are redistributed only for papers under Creative Commons licences, with their original licences; all others are linked.

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